

Do Stock Returns in India Follow a Random Walk?

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This paper empirically investigates the behavior of stock returns of two premier stock markets in India, namely, the Bombay Stock Exchange (BSE) and National Stock Exchange (NSE). Specifically, the paper seeks to examine whether the security returns in these two markets follow Random Walk Hypothesis (RWH). Towards this end, data on major indexes during the period June 2, 1997 to February 29, 2009 are analyzed using variance ratio and autocorrelation tests. The results are inconclusive as both the tests provide mixed results.

Introduction

The Random Walk Hypothesis (RWH) has been extensively tested in the sphere of financial economics over the past decades (Fama, 1965; and Lo and MacKinlay, 1988). It asserts that returns of financial securities are generated by a random process so that it is not possible to predict their future movements based on past information. In this study an attempt is made to empirically investigate whether RWH holds or not in the context of two premier Indian stock markets, namely, Bombay Stock Exchange (BSE) and National Stock Exchange (NSE).

Though there are a few studies on Indian stock markets (Sharma and Kennedy, 1977; Poshakwale, 2002; and Chaudhuri and Wu, 2003a and 2003b), this study is justified on the following grounds. First, the available studies refer to the 1980s and 1990s. Since the market microstructure of Indian stock markets has undergone various changes in the recent past owing to measures such as introduction of derivatives,¹ a study covering the period of such structural changes is in order. Second, most of the recent studies (e.g., Chaudhuri and Wu, 2003a and 2003b) are based on monthly data, while the present study has used data of daily frequency. Finally, the study covers two premier stock markets in India, namely, BSE and NSE. It covers eight indexes from the NSE and six indexes from the BSE. We have employed parametric tests such as variance ratio test propounded by Lo and MacKinlay (1988), and autocorrelation test using Ljung-Box Q -statistics, to investigate the behavior of asset returns.

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¹ Index futures and options were introduced in June 2000 and June 2001 respectively, and futures and options on individual stocks were introduced in July 2001 and November 2001 respectively.

The rest of the paper is organized as follows: it briefly reviews the previous works on RWH, followed by an outline of the data and methodology used in the study. Then the empirical results are presented, and finally, the conclusion is offered.

Review of Literature

The previous literature on RWH is vast and the results are conflicting. Therefore, we attempt to present only a brief review of the relevant works in this area. Sharma and Kennedy (1977), using runs test and spectral technique, find that monthly returns in the BSE follow RWH. The studies of Working (1960), Samuelson (1965), Fama (1965), Niederhoffer and Osborne (1966), and Fama (1970) suggest that stock price movements are not serially correlated and hence it is impossible to make abnormal profits from random investment. Similar results are reported by Jennergren and Korsvold (1974) in their study of Norwegian and Swedish markets. Contrary to these findings, French and Roll (1986) observe statistically significant negative serial correlation in daily returns even though they were skeptical about the economic significance of such returns. In a similar vein, Keim and Stambaugh (1986) find statistically significant predictability in stock prices by using forecasts of predetermined variables. Interestingly, Fama and French (1988) cast doubts on the validity of RWH showing that long-horizon returns are negatively correlated. In the same year, Lo and MacKinlay (1988), using variance ratio test, proved that stock returns do not follow RWH. While the study of Poterba and Summers (1989) reveal positive serial correlation for short, and negative serial correlation for long-horizon returns, Jagadeesh (1990) reports positive serial correlation for long horizons.

The results of studies on the RWH from emerging markets are also inconclusive. Consistent with the observation of Harvey (1993) that emerging markets do not follow RWH as they are less efficient than their developed counterparts, the studies by Butler and Malikah (1992), Frennberg and Hansson (1993), Ayadi and Pyun (1994), and Huang (1995) also reject RWH for different markets like Saudi Arabia, Kuwait, Sweden and Korea. While Urrutia (1995) finds positive autocorrelation in monthly returns of some Latin American countries, Liu *et al.* (1997) report that Chinese markets are efficient. Two other studies on Latin American stock markets by Ojah and Karemera (1999) and Greib and Reyes (1999) provide mixed results. While the former study finds evidences in favor of RWH for Latin America, the latter finds evidences against RWH in the Mexican market and pro-RWH results in the Brazilian market. Along the same line, Balver *et al.* (2000) note the rejection of RWH in 18 developed nations.

Refuting the findings of previous studies, Darrat and Zhong (2000) and Lee *et al.* (2001) provide ample evidence in favor of market efficiency. Consistent with previous studies, Ryoo and Smith (2002) report that Korean market returns do not follow the RWH.

Abraham *et al.* (2002) show that observed index levels in Saudi Arabia, Kuwait and Bahrain do not confirm to RWH, whereas the corrected returns are supportive of weak-form of market efficiency. Out of a total of seven countries investigated by Smith *et al.* (2002), six markets do not follow RWH, except South African market where empirical evidences are in favor of RWH. Poshakwale (2002), Chaudhuri and Wu (2003a) and Smith and Ryoo (2003), by and large, find evidences against RWH in different markets like India, 17 emerging economies and four European markets, respectively. Rejection of random walk from 17 emerging Asian markets including India is shown by Chaudhuri and Wu (2003b).

Very recently, Hoque *et al.* (2007) also find rejection of RWH in majority of eight emerging markets. Nakamura and Small (2007), by employing a new method, Small Shuffle Surrogate, (SSS) conclude that the US and Japanese markets follow RWH. Similar are the findings of Lock (2007) on Taiwan market. To sum up, the review of literature provides a mixed picture of empirical evidences on RWH.

Data

This study used daily sample data on 14 indexes of varying time ranges from BSE and NSE. The sample indexes and their various time ranges are given in the Appendix. We have considered these indexes, as together they are expected to represent the market. Moreover, most of these indexes have the track record of at least five years. The index values of NSE are collected from the official website of NSE, and those of BSE are collected from the CMIE-PROWESS database.

Methodology

As the foregoing review of literature reveals, empirical analysis of RWH has been conducted by using various techniques. In the present study, we employ parametric tests, namely, variance ratio and autocorrelation tests. These tests have been popular in the literature. We briefly describe the two tests.

Variance Ratio Test

The variance ratio test proposed by Lo and MacKinlay (1988) is capable of distinguishing among several interesting alternative stochastic processes. It exploits the fact that the variance of the increments of a random walk is linear in the sampling interval. For, example, if the stock prices are generated by a random walk process, then the variance of monthly sampled log-price relatives must be four times as large as the variance of weekly return. Therefore, if we obtain $nq + 1$ index values, $I_0, I_1, I_2, \dots, I_{nq}$ at equally spaced intervals, where q is any integer greater than 1, the ratio of $1/q$ of the variance $I_t - I_{t-q}$ to the variance of $I_t - I_{t-1}$ would be equal to 1. The null hypothesis is that the variance ratio is equal to 1. The variance ratio, $VR(q)$, is defined as:

$$VR(q) = \frac{\sigma^2(q)}{\sigma^2(1)} \quad \dots(1)$$

where $\sigma^2(q)$ is $1/q$ times the variance of the q differences and $\sigma^2(1)$ is the variance of the first differences. They are calculated as given by Equations (2) and (3) respectively:

$$\sigma^2(q) = \frac{1}{m} \sum_{t=q}^{nq} (I_t - I_{t-q} - q\hat{\mu})^2 \quad \dots(2)$$

where

$$m = q(nq - q + 1) \left(1 - \frac{q}{nq}\right)$$

and

$$\sigma^2(1) = \frac{1}{(nq - 1)} \sum_{t=1}^{nq} (I_t - I_{t-1} - \hat{\mu})^2 \quad \dots(3)$$

where

$$\hat{\mu} = \frac{1}{nq} (I_{nq} - I_0)$$

We follow heteroscedasticity-consistent standard normal test that uses overlapping interval. The test statistic, $Z^*(q)$, is as follows:

$$Z^*(q) = \frac{VR(q) - 1}{[\phi^*(q)]^{1/2}} \quad \dots(4)$$

where

$$\phi^*(q) = \sum_{j=1}^{q-1} [2(q-j)/q]^2 \hat{\delta}(j)$$

where

$$\hat{\delta}(j) = \frac{\sum_{t=j+1}^{nq} (I_t - I_{t-1} - \hat{\mu})^2 (I_{t-j} - I_{t-j-1} - \hat{\mu})^2}{\left[\sum_{t=1}^{nq} (I_t - I_{t-1} - \hat{\mu})^2 \right]^2}$$

Autocorrelation Test

Autocorrelation estimates may be used to test the hypothesis that the process generating the observed return is a series of independently and identically distributed (iid) random variables. It helps to evaluate whether successive values of serial correlation is significantly different from zero. The autocorrelation function (ρ_k) of a series (I_t) at lag k is given by:

$$\rho_k = \frac{\sum_{t=k+1}^T (I_t - \mu)(I_{t-k} - \mu)}{\sum_{t=1}^T (I_t - \mu)^2}$$

where μ denotes sample mean of the return. To test the joint hypothesis that all autocorrelation coefficients are simultaneously equal to zero, we have used the Ljung-Box portmanteau Q -statistic. If the null hypothesis is rejected, then the underlying series is expected to follow a random walk.

Empirical Analysis

The descriptive statistics of daily index returns are given in Table 1. It is evident that mean return for all indexes is very low and CNX IT has the highest standard deviation (0.052). The skewness and kurtosis are high for all index returns and negative skewness implies that the series are flatter to the left compared to the normal distribution. The kurtosis indicates that return distribution has sharp peaks compared to a normal distribution. Similarly, Jarque-Bera statistic also shows that index return is not normally distributed.

The empirical results of the variance ratio tests are furnished in Table 2. The variance ratio and the corresponding homoscedasticity-robust test statistic for various investment horizons like 2, 4, 8 and 16 days are computed. As per the variance ratio test, under the null hypothesis of random walk, the ratio is expected to be equal to 1. Our results are mixed. Indexes like CNX IT and BSE 200 are found to be following random walk at all investment horizons. Similarly, CNX Nifty, CNX Infrastructure, and BSE Sensex also follow random walk except at the first investment horizon. In contrast, CNX Nifty Junior, BSE 100, BSE 500, BSE Midcap and BSE Small Cap completely reject the RWH at all investment horizons. CNX 500 seems to suggest rejection of random walk except at first horizon of investment.

The autocorrelation test results are presented in Table 3. It is evident from the Table that CNX Nifty Junior and BSE 500 indicate random walk, as we cannot reject the null hypothesis of no serial correlation as the p -value is close to 1. This implies that in the case of other indexes, random walk is rejected since the series are correlated. Similarly, as we have seen in the variance ratio test, first-order serial correlation coefficients of all index returns are highly statistically significant.

Table: 1: Descriptive Statistics of Stock Returns									
Variable	Mean	Median	Minimum	Maximum	SD	Skewness	Kurtosis	Jarque-Bera	p-Value
CNX Nifty	0.000340	0.001168	-0.130539	0.079691	0.017438	-0.519211	4.475148	2566.928	0.000000
CNX Nifty Junior	0.000456	0.001895	-0.131333	0.082922	0.020527	-0.678203	3.807871	1987.318	0.000000
CNX 100	0.000654	0.002108	-0.130494	0.080065	0.017992	-0.859112	5.926253	2414.447	0.000000
CNX 500	0.000427	0.001904	-0.128847	0.076945	0.017731	-0.772766	4.553660	2327.864	0.0000
CNX Defty	0.000233	0.001184	-0.141131	0.089858	0.018438	-0.472041	4.698742	2792.704	0.0000
CNX IT	0.000175	0.000998	-2.365839	0.145567	0.052223	-32.051150	1449.945	2327.864	0.0000
CNX Bank	0.000657	0.000735	-0.151380	0.114014	0.021577	-0.419269	4.193256	1731.887	0.0000
CNX Infrastructure	0.000676	0.002033	-0.150214	0.102127	0.021883	-0.767701	6.057552	2064.842	0.0000
BSE Sensex	0.000342	0.001147	-0.118092	0.085915	0.017754	-0.371325	3.511066	1471.438	0.0000
BSE 100	0.000395	0.001556	-1.473311	0.552934	0.023958	-1.473311	244.0782	66926.37	0.0000
BSE 200	0.000407	0.033690	-1.037087	1.084561	0.033690	1.394998	712.4875	5085.416	0.0000
BSE 500	0.000255	0.001944	-0.249828	0.075327	0.018640	-1.727030	17.37390	2973.094	0.0000
BSE Mid Cap	0.000145	0.002699	-0.120764	0.078359	0.017902	-1.373015	7.010821	3002.329	0.0000
BSE Small Cap	0.000233	0.002825	-0.108357	0.064767	0.018712	-1.234866	4.141240	1232.220	0.0000

Table 2: Variance Ratio Estimates of Daily Index Values				
Index	Number q of Base Observations Aggregated to Form Variance Ratio			
	2	4	8	16
CNX Nifty	1.062 (3.35)*	1.053 (1.5)	1.036 (0.6)	1.087 (1.0)
CNX Nifty Junior	1.143 (7.7)*	1.209 (6.0)*	1.231 (4.2)*	1.404 (4.9)*
CNX 100	1.093 (3.6)*	1.096 (2.0)*	1.054 (0.7)	1.126 (1.1)
CNX 500	1.138 (6.8)	1.189 (4.9)*	1.221 (3.6)*	1.380 (4.2)*
CNX Defty	1.072 (3.8)*	1.094 (2.7)*	1.091 (1.6)	1.163 (2.0)*
CNX IT	1.008 (0.43)	1.016 (0.44)	1.026 (0.48)	1.113 (1.3)
CNX Bank	1.123 (5.9)*	1.146 (3.7)*	1.049 (0.80)	1.047 (0.51)
CNX Infrastructure	1.091 (3.2)*	1.078 (1.49)	1.031 (0.38)	1.061 (0.49)
BSE Sensex	1.070 (3.6)*	1.069 (1.9)	1.034 (0.6)	1.093 (1.1)
BSE 100	0.840 (−8.3)*	0.769 (−6.4)*	0.719 (−4.9)*	0.770 (−2.7)*
BSE 200	1.011 (0.61)	1.014 (0.4)	1.023 (0.42)	1.058 (0.6)
BSE 500	1.123 (5.9)*	1.173 (4.4)*	1.217 (3.5)*	1.396 (4.2)*
BSE Mid Cap	1.220 (7.8)*	1.350 (6.6)*	1.464 (5.5)*	1.688 (5.5)*
BSE Small Cap	1.279 (9.9)*	1.504 (9.6)*	1.733 (8.8)*	2.069 (8.6)*
Note: The variance ratios $VR(q)$ are reported in the main rows and heteroscedasticity-robust test statistics $Z^*(q)$ are given in parentheses. Under the random walk hypothesis, the value of $VR(q)$ is 1. Asterisk values reject random walk hypothesis at 5% level of significance.				

Table 3: Autocorrelation Test Results			
Index	Lags	LB Q-Statistic	Q Significance
CNX Bank	15	73.78	0.0000
CNX 100	15	39.40	0.0006
CNX 500	15	81.13	0.0000
CNX Defty	15	91.83	0.0004

Table 3 (Cont.)

Index	Lags	LB Q -Statistic	Q Significance
CNX Infrastructure	15	40.06	0.0004
CNX IT	15	43.05	0.0002
CNX Nifty Junior	15	17.05	0.9156
CNX NIFTY	15	124.89	0.0000
BSE 100	15	41.82	0.0002
BSE 200	15	92.64	0.0000
BSE 500	15	5.790	0.9831
BSE Mid cap	15	59.73	0.0000
BSE Small cap	15	96.27	0.0000
BSE Sensex	15	149.22	0.0000
Note: The LB Q -statistic is given in the table at lag 15 for all series. The test statistic rejects the null hypothesis of no serial correlation at all conventional significance levels, except for CNX Nifty Junior and BSE 500.			

Conclusion

This study investigated whether stock returns in India follow a RWH or not. Two parametric tests, namely, variance ratio of Lo and MacKinlay (1988) and autocorrelation tests, were used to analyze the daily data on 14 market indexes from the two major stock exchanges in India, i.e., BSE and NSE. The variance ratio test provides mixed results of the behavior of returns of different indexes. The autocorrelation test broadly rejects the RWH for all indexes, except Nifty Junior and BSE 500. In other words, the autocorrelation test finds evidences against random walk in most of those indexes, for which variance ratio test found evidences in favor of random walk.

Even though the results of the empirical analysis are mixed, evidences against RWH from variance ratio test seem to be strong, as the variance ratio test is known to be more powerful than the autocorrelation test. The empirical results of this study appear to be sensitive to the tests employed for the analysis. Given such a fact, the results of the study corroborate Fama's (1998) stand that the choice of methodology of the analysis of return process might influence the results and in the long run the market remains efficient. ♦

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Appendix

Sample Indexes – Time Period Covered			
S. No.	Index	Time Period	% to Total Market Capitalization
1.	CNX Nifty	02/06/1997-30/01/2009	61.70
2.	CNX Nifty Junior	02/06/1997-30/01/2009	9.89
3.	CNX 100	01/01/2003-30/01/2009	68
4.	CNX 500	07/06/1999-30/01/2009	84.24
5.	CNX Defty	02/06/1997-30/01/2009	–
6.	CNX IT	02/06/1997-30/01/2009	14
7.	CNX Bank	01/01/2000-30/01/2009	7
8.	CNX Infrastructure	01/01/2004-30/01/2009	18.84
9.	BSE Sensex	01/01/1998-30/01/2009	46.53
10.	BSE 100	01/01/1998-30/01/2009	75.67
11.	BSE 200	01/01/1998-30/01/2009	85.24
12.	BSE 500	03/01/2000-30/01/2009	93.51
13.	BSE Mid Cap	01/01/2004-30/01/2009	12.80
14.	BSE Small Cap	01/01/2004-30/01/2009	3.74
Note: The values in the last column are latest values available in the website of BSE and NSE.			

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